Venkata Naga Sai Bhavan Koppolu

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Research Interests

Research interests in embodied intelligence and robot learning, with focus on vision-language-action frameworks, reinforcement learning, and foundation models for autonomous physical agents operating in unstructured environments. Broader interests include sim-to-real transfer, imitation learning and offline RL from human demonstrations, multi-task and meta-learning for generalizable robot policies, uncertainty-aware perception for safe navigation and safe human-robot interaction.

Education

Arizona State University — Tempe, AZ
M.S. in Robotics and Autonomous Systems, Artificial Intelligence — *Jan. 2025 – Dec. 2026*

Queen Mary University of London — London, UK
B.S. in Computer Science — *Sept. 2021 – May 2024*

Publications

Uncertainty-Aware Lightweight Terrain Segmentation for Risk-Aware Rover Navigation 2026

- Venkata Naga Sai Bhavan Koppolu. Uncertainty-aware lightweight neural network for terrain classification and segmentation, enabling risk-aware navigation decisions for planetary rovers under strict compute and energy constraints. *Manuscript in preparation for submission to ICRA, IROS.*

Sentiment-Aware Fake Review Detection in Dravidian Languages 2026

- Venkata Naga Sai Bhavan Koppolu. Cross-lingual fake review detection across Telugu, Tamil, and Kannada using adversarially generated reviews and multilingual transformer models. *Manuscript in preparation for submission to ACL, EMNLP.*

Awards & Honors

1st Place, AWS DeepRacer Contest | Arizona State University — Apr. 2026

4th Place, Top 10 Most Innovative Undergraduate Projects | Queen Mary University of London — May 2024

Certifications

Fundamentals of Engineering Project Management | Ira A. Fulton Schools of Engineering — Aug. 2026

Inter-Personal Skills in Engineering Project Management | Ira A. Fulton Schools of Engineering — Aug. 2026

Projects

Uncertainty-Aware Martian Terrain Segmentation Jan. 2026 - Present

- Extending prior terrain classification (Tiny CNN, MobileNetV2, SNN on the 16,064-image AI4MARS dataset, 83.3% accuracy) to pixel-level semantic segmentation using a lightweight encoder-decoder architecture optimized for rover-class hardware.
- Integrating Monte Carlo Dropout and deep ensembles to produce calibrated uncertainty maps, decomposing prediction confidence into epistemic and aleatoric components per terrain class.
- Applying uncertainty estimates to support risk-aware rover path planning, flagging high-uncertainty regions as traversal hazards; validating inference feasibility under strict edge compute and power budgets.

Sentiment-Aware Fake Review Detection in Dravidian Languages Sept. 2023 - Present

- Engineered a low-resource NLP dataset by translating the 1,600-review OpSpam corpus into Telugu; fine-tuned mBERT achieving 79.1% accuracy vs. 85.7% English baseline, establishing the first cross-lingual deception detection benchmark for Telugu.
- Benchmarked Zero-Shot transfer (61.6%) and Mixed-Language Training at varying composition ratios, with 50/50 achieving 75.9%, quantifying the performance–data-efficiency trade-off for low-resource multilingual NLP.
- Expanding scope to Tamil and Kannada via DravidianLangTech shared task; evaluating instruction-tuned LLMs (Llama 3, Gemma) as zero/few-shot baselines against fine-tuned multilingual encoders and investigating back-translation augmentation for data-scarce settings.

Multi-Robot Coordination for Martian Terrain Exploration Aug. 2026 - Present

- Designing a decentralized multi-rover coordination framework for autonomous Martian surface exploration under intermittent communication constraints.
- Implementing consensus-based task allocation and distributed path planning algorithms for heterogeneous robot teams navigating uncertain terrain; evaluating coordination performance and scalability across 3–10 agent configurations.
- Integrating semantic terrain maps derived from prior AI4MARS classification work as shared environment representations, enabling informed inter-rover coverage planning and hazard-aware task assignment.

Hindsight Experience Replay: Reproducing Goal-Conditioned RL Aug. 2026 - Present

- Conducting a reproduction study of Andrychowicz et al.'s Hindsight Experience Replay (HER) algorithm combined with DDPG on OpenAI Gym Fetch robotic manipulation tasks (Reach, Push, PickAndPlace, Slide).
- Designed and evaluated experiments comparing goal-relabeling strategies (future, episode, random), analyzing their effects on sample efficiency, convergence rate, and policy stability across sparse-reward task variants.
- Implemented full training, evaluation, and logging pipelines in PyTorch with MuJoCo physics simulation, tracking reward curves and success rates across relabeling strategies to validate findings against the original paper.
- Analyzing failure modes in sim-to-real transfer, identifying policy brittleness under observation noise and contact dynamics mismatch as key bottlenecks for deployment on physical manipulators.

Memorization in Diffusion Models Jan. 2026 - Apr. 2026

- Conducted an experimental study on memorization behavior in diffusion models using a simplified DDPM U-Net architecture trained on CIFAR-10.
- Designed and evaluated experiments analyzing the effects of conditioning type, dataset size, training duration, and label specificity on memorization behavior.
- Implemented diffusion training, sampling, and memorization evaluation pipelines in PyTorch, utilizing A100 GPUs across Google Colab and the ASU Sol Supercomputer, reproducing memorization rates of up to 99.8% under controlled conditions.

Identifying LLM Performance Gaps Across Languages Aug. 2025 - Dec. 2025

- Evaluated OLMo-2-7B-Instruct across 100+ languages and 10+ benchmarks using vLLM and HPC infrastructure, establishing multilingual performance baselines.
- Built a multilingual translation pipeline using NLLB-200 and Hunyuan-MT-7B, validating outputs with BLEU, METEOR, COMET, and LaBSE metrics.
- Quantified token exposure vs. accuracy (Spearman's ρ of 0.94), identifying data imbalance as the main driver of multilingual performance gaps.

Agentic AI Robotic Puzzle-Solver Aug. 2025 - Dec. 2025

- Integrated perception with planning and control layers by developing an autonomous Vision-Language-Action (VLA) framework using Google Gemini 2.5 Flash for a physical robotic manipulator.
- Designed semantic mapping and environment representations, utilizing 3x3 Affine Transformations to map 2D OpenCV bounding boxes into depth-aware 3D spatial coordinates.
- Implemented LLM function calling to abstract complex spatial reasoning into Python motion controls, mitigating real-world constraints like visual hallucinations to ensure deterministic kinematic execution.

AWS DeepRacer – Reinforcement Learning Jan. 2026 - Apr. 2026

- Engineered a geometry-aware reward function with curvature-sensitive speed shaping, applying policy gradient principles to replace a naive centering reward that caused instability and suboptimal convergence on tight turns.
- Redesigned the discrete action space to decouple speed and steering, reducing state-action coupling and improving exploration efficiency across the policy's decision boundary.
- Applied curriculum learning and hyperparameter tuning across progressively harder track configurations, achieving significant lap time reduction and first-place finish in the final contest evaluation.

FDM Wellbeing App | Client: FDM Group (UK) Jan. 2023 - Apr. 2023

- Engineered CI/CD test automation pipelines for an enterprise-grade web application using ReactJS and Node.js, systematically validating 6+ complex software workflows.
- Architected and verified backend database infrastructure by testing and linking 25+ API endpoints, ensuring continuous delivery across the development lifecycle.
- Improved end-to-end system efficiency by deploying automated acceptance testing across 15+ core use cases, eliminating flaky builds and validating predictable performance.

Professional Experience

AI/ML Systems Engineering Intern Dec. 2025 - Aug. 2026
Innominds *Remote*

- Engaged directly with Fortune 100 clients to scope, develop, and present AI and automation solutions, translating complex technical architectures into actionable business use cases and POC deliverables.
- Built an Agentic BI POC for Novartis combining a NetworkX/Pyvis knowledge graph, an autonomous pipeline monitoring engine, and an NLP query interface (Llama 3.2, LangChain) over synthetic molecular R&D data; delivered live Colab demo to enterprise stakeholders.
- Authored detailed AI annotation pipeline specifications for a Qualcomm autonomous driving dataset, defining transformer-based 3D LiDAR pre-annotation, multi-sensor calibration validation, confidence-based human-in-the-loop routing, and versioned ground truth workflows.

Software Engineering Intern June 2023 - Sept. 2023
Techwish *Hyderabad, India*

- Developed backend microservices using Java, Spring Boot, Hibernate, and MySQL for live projects serving Fortune 100 clients.
- Built responsive HTML/CSS frontend components and integrated REST APIs to support user-facing functionality
- Developed JUnit test cases, used Git/GitHub version control, and contributed across the Software Development Life Cycle (SDLC).

Software Engineering Intern June 2022 - Sept. 2022
Innominds *Hyderabad, India*

- Developed full-stack features across Java, .NET, and ASP.NET codebases on live projects under senior Technology Leaders.
- Built responsive HTML/CSS UI components to improve frontend functionality and backend service integration.
- Contributed to Agile sprints through requirements gathering, client interactions, and technical documentation.

Academic Experience

Course Grader / Teaching Assistant Aug. 2025 - Present
Arizona State University *Tempe, AZ*

- **Managing Intelligent Devices in an Enterprise Environment**: supported lab-focused instruction on IoT ecosystem architecture, single-board computer programming (Raspberry Pi), machine-to-machine networking, cloud-based IoT management (AWS), and IoT security; graded Python programming assignments and technical reports.
- **AI into Cybersecurity for IT**: graded coursework covering AI/ML terminology and frameworks (deep learning, NLP, neural networks), practical cybersecurity use cases, AI-driven threat detection, anomaly detection, and adversarial AI; course draws on hands-on AI cybersecurity tools and algorithms.
- **Wind Energy**: graded problem sets and lab reports on wind turbine systems, energy conversion, and grid integration.

Technical Skills

Languages: Python, Java, JavaScript, C++, R, SQL, MATLAB, Bash
AI, ML & Generative AI: PyTorch, TensorFlow, Hugging Face, JAX, scikit-learn, Supervised Learning, Fine-Tuning, RAG, Agentic AI, Quantization, LLM/vLLM, LangChain, Diffusion Models, RLHF, Instruction Tuning, Monte Carlo Dropout, Deep Ensembles, Uncertainty Estimation
NLP & Computer Vision: Multilingual NLP, Transformers, Text Classification, Object Detection, Image Segmentation, Semantic Segmentation, Neural Networks, OpenCV, Sentiment Analysis, Named Entity Recognition
Robotics & Autonomous Systems: ROS/ROS2, Motion Planning, Inverse Kinematics, Perception–Planning–Action Pipelines, Reinforcement Learning, DDPG, HER, Goal-Conditioned RL, Multi-Robot Coordination, Consensus Algorithms, Distributed Path Planning, Sensor Fusion, PID Control, OpenAI Gym (MuJoCo/Fetch), PyBullet, Isaac Sim, Gazebo, AWS DeepRacer, Neuromorphic Computing
Cloud, HPC & MLOps: AWS, GCP, Azure, Docker, Kubernetes, SLURM, HPC/Supercomputing, Distributed Training, MLflow, Model Parallelism, Weights & Biases, CI/CD
Web, Data & Tools: React, Node.js, Spring Boot, APIs, NumPy, Pandas, Matplotlib, MySQL, Git/GitHub, Linux

Spoken Languages

Fluent: English, Telugu, Hindi
Intermediate: German
Elementary: French, Tamil

Activities

Sun Devil Gaming | Arizona State University — Aug. 2025 - Present
Queen Mary Gaming Society | Queen Mary University of London — Jan. 2022 - Apr. 2024
Football Team Captain | Kennedy High the Global School — Jun. 2017 - Mar. 2020
Model United Nations Speaker | Kennedy High the Global School — July 2020

References

Dr. Haim Dubossarsky | Professor, Queen Mary University of London
- Supervisor, NLP-Based Fake Review Detection in Telugu
- h.dubossarsky@qmul.ac.uk

Ronald Calhoun | Assistant Teaching Professor, Arizona State University
- Supervisor, Uncertainty-Aware Lightweight Terrain Segmentation for Risk-Aware Rover Navigation
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